

Prangon Barua

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EDUCATION

Burlington High School
High School Diploma

Aug 2024 – Jun 2028
Burlington, MA

EXPERIENCE

BOOST Intern

Apr 2026 – Present

MassRobotics

Boston, MA

- Selected for MassRobotics' BOOST program, an immersive cohort connecting students with leading Boston-area robotics companies and founders
- Toured Autodesk's facilities and met with engineers and product teams to explore real-world applications of design and manufacturing technology
- Built relationships with founders and engineers across multiple early-stage robotics startups, gaining direct exposure to hardware development and go-to-market strategy
- Developing an independent engineering project to be presented to a panel of startup founders at program's conclusion

FOCUS Engineering Intern

Sep 2025 – Dec 2025

BAE Systems, Inc.

Nashua, NH

- Collaborated with BAE engineers to design and build a functional robot with custom software over a three-month sprint
- Integrated software, electrical, and mechanical subsystems through cross-disciplinary teamwork with student and professional engineers
- Presented the final project to an audience of ~50 attendees, including 15 BAE engineers, highlighting the technical approach and impact

Drone Academy Participant

Jul 2025

MassRobotics

Boston, MA

- Selected for a competitive cohort focused on UAV design, fabrication, and flight operations
- Built and tested drones hands-on, developing skills in soldering, circuit assembly, and flight tuning
- Networked with engineers from Amazon Robotics, MITRE, Microsoft, and MIT; presented project to the Mayor of Boston

BlueTech Academy Participant

Jul 2025

MassRobotics (with MITRE & Greensight)

Boston, MA

- Designed and prototyped two autonomous underwater vehicles (AUVs), a silicone AUV and starfish-model platform, with MITRE and Greensight engineers
- Integrated Arduino-based control circuits with custom sensors and actuators, gaining hands-on experience in PCB integration and underwater robotics

Practical Radio Intern

Mar 2025 – Apr 2025

MIT Beaver Works Summer Institute (BWSI)

Cambridge, MA

- Collaborated with MIT students to design and prototype antennas and oscillators for real-world communication systems
- Established a successful long-distance contact with Case Western Reserve University using MIT's official amateur radio call sign **W1MX**
- Iterated on hardware designs based on peer feedback, deepening practical knowledge in RF and signal transmission

PROJECTS

- EloEats** | *Next.js, TypeScript, Claude Vision, Tailwind* May 2026
- Built a gamified healthy-eating app that scans fridge contents, scores items nutritionally, and ranks users on a live leaderboard
 - Implemented camera-based inventory scans and “log consumed” diffs using Claude Vision for item recognition
 - Co-built with **Ariel Orlov**; shipped a hackathon proof-of-concept with demo mode and responsive UI
- Alphex** | *Next.js, TypeScript, FastAPI, Supabase, pgvector* Apr 2026
- Built a research platform that imports papers (PDF, arXiv, DOI, BibTeX) and indexes them for semantic search
 - Implemented streaming RAG chat with grounded citations and tools for literature review, comparison, and question generation
 - Delivered L^AT_EX export for papers/notes and deployed a full-stack web app with Next.js + FastAPI
- Relate** | *React Native, Expo, TypeScript, Express, SQLite, Studio, RAG* Mar 2026
- Built a free, multilingual mobile app for immigrant parents of autistic children, available in **12 languages** via i18next with instant in-app switching
 - Architected a React Native frontend (Expo Router, NativeWind, Zustand) across five tabs and a TypeScript Express backend with JWT/bcrypt auth and SQLite storage seeded with 57 skills and 42 resources
 - Integrated a local Qwen3-14b model via LM Studio with a custom RAG pipeline over 15 original knowledge files covering IEPs, AAC, school rights, and legal protections
- Oscillator Prototype** | *Circuit Design, Vacuum Tube, Signal Processing* Mar 2025
- Designed and built an oscillator at MIT BWSI using a vacuum tube for amplification, potentiometer for frequency control, and resistor/capacitor networks for signal stabilization
 - Troubleshooted and refined the system in collaboration with MIT students and alumni to achieve audio signal output for communication
- Yagi Antenna** | *RF Engineering, Antenna Design, Soldering* Mar 2025
- Designed and assembled a directional Yagi antenna at MIT BWSI in collaboration with MIT students: wooden boom with metal director/reflector elements spaced for resonance, fed via coaxial cable to a handheld receiver
 - Hand-soldered all element-to-feed connections and tuned element spacing to optimize directional gain
 - Field-tested the build in a team-based foxhunt across MIT’s campus, using two antennas and radios per team to triangulate and locate a hidden transmitter
- Music Industry Analytics Tool** | *Python, Data Analysis* Jan 2025
- Built a Flask web app that analyzes Billboard Hot 100 chart history by artist and exports results to Excel

HONORS & AWARDS

- 2nd Place, Dream Venture Hackathon** (May 2026) — Issued by the Dream Venture Team. Business and technology hackathon for high school students; developed a 2nd-place project with software to solve a real-world problem.
- 4th Place, Special Competitive Studies Hackathon @ Harvard** (Apr 2026) – Placed 4th out of a field of undergraduates and PhD students with a research-acceleration project, competing against entrants with prior industry experience

CERTIFICATIONS

- Python Core 2026** – MIT Lincoln Laboratory (Mar 2026)
- Computer Science and Systems 2026** – MIT Lincoln Laboratory (Mar 2026)
- Google AI Essentials** – (May 2025)
- CS50** – Harvard University (Dec 2024)

TECHNICAL SKILLS

- Languages:** Python, C/C++, JavaScript, TypeScript, SQL, Java, Git, L^AT_EX
- Frameworks & Libraries:** React Native, Expo, Express, Node.js, NativeWind, Zustand
- Engineering & Hardware:** Electrical Engineering, Circuit Design, Arduino, RF Engineering, PCB & Oscillator Design, UAV / AUV Prototyping, CNC machining, soldering, laser cutting
- Tools & Platforms:** Git, LM Studio, SQLite, REST APIs, RAG, JWT/bcrypt